Database Design Document

AssetFlow AI

Database Architecture

Version: 1.0

Database: PostgreSQL

ORM: Prisma

Table of Contents

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1. Database Overview

AssetFlow AI uses a relational PostgreSQL database.

Goals

- Fast Search
- High Integrity
- ACID Transactions
- Easy Reporting
- AI-ready Data

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2. Entity Relationship Diagram

```text
Departments
      |
      | 1
      |
      ▼
Users
      |
      |1
      ▼
Assignments
      |
      |Many
      ▼
Assets
      |
```

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▼ ▼

Maintenance AssetHistory

|

▼

Notifications

Assets

|

▼

Invoices

Vendors

|

▼

Assets
```

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# 3. Naming Convention

Table Names

snake_case

Example

users

assets

maintenance
```

Primary Key

id

Foreign Keys

user_id

asset_id

department_id

Timestamps

created_at

updated_at

deleted_at

---

4. Tables

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users

Purpose

Stores all system users.

Columns

Column	Type
id	UUID
full_name	VARCHAR(100)
email	VARCHAR(255)
password_hash	TEXT
role	ENUM
phone	VARCHAR(20)
department_id	UUID
profile_image	TEXT
status	BOOLEAN
created_at	TIMESTAMP
updated_at	TIMESTAMP
deleted_at	TIMESTAMP

---

departments

Column	Type
id	UUID
name	VARCHAR
description	TEXT
manager_id	UUID
location	VARCHAR
created_at	TIMESTAMP
updated_at	TIMESTAMP

---

vendors

Column	Type
id	UUID
vendor_name	VARCHAR
email	VARCHAR
phone	VARCHAR
address	TEXT
gst_number	VARCHAR
created_at	TIMESTAMP

---

assets

Purpose

Stores every asset.

Column	Type
id	UUID
asset_code	VARCHAR
asset_name	VARCHAR
category	VARCHAR
serial_number	VARCHAR
model	VARCHAR
manufacturer	VARCHAR

purchase_date	DATE
purchase_price	DECIMAL
warranty_expiry	DATE
condition	ENUM
status	ENUM
location	VARCHAR
department_id	UUID
vendor_id	UUID
qr_code	TEXT
image_url	TEXT
created_at	TIMESTAMP
updated_at	TIMESTAMP
deleted_at	TIMESTAMP

---

assignments

Tracks which employee currently owns an asset.

Column	Type
id	UUID
asset_id	UUID
user_id	UUID
assigned_by	UUID
assigned_date	DATE
expected_return	DATE
returned_date	DATE
remarks	TEXT

---

maintenance

Column	Type
id	UUID
asset_id	UUID
technician_id	UUID
issue	TEXT
priority	ENUM
status	ENUM
repair_cost	DECIMAL
opened_at	TIMESTAMP
completed_at	TIMESTAMP

---

asset_history

Stores complete lifecycle.

Column	Type
id	UUID
asset_id	UUID
action	VARCHAR
user_id	UUID
description	TEXT
created_at	TIMESTAMP

Example Actions

- Created
- Assigned
- Returned
- Maintenance Started
- Maintenance Completed
- Disposed

---

invoices

Column	Type
id	UUID
asset_id	UUID
invoice_number	VARCHAR
invoice_date	DATE
amount	DECIMAL
pdf_url	TEXT

---

notifications

Column	Type
id	UUID
user_id	UUID
title	VARCHAR

message	TEXT
is_read	BOOLEAN
created_at	TIMESTAMP

---

audit_logs

Tracks every sensitive activity.

Column	Type
id	UUID
user_id	UUID
action	VARCHAR
table_name	VARCHAR
record_id	UUID
old_value	JSONB
new_value	JSONB
ip_address	VARCHAR
created_at	TIMESTAMP

---

5. Relationships

users

↓

belongs to

↓

departments

---

assets

↓

belongs to

↓

vendors

---

assets

↓

belongs to

↓

departments

---

assignments

↓

belongs to

↓

users

---

assignments

↓

belongs to

↓

assets

---

maintenance

↓

belongs to

↓

assets

---

maintenance

↓

belongs to

↓

technicians(users)

---

asset_history

↓

belongs to

↓

assets

---

notifications

↓

belongs to

↓

users

---

6. Enums

Role

```text
ADMIN
MANAGER
TECHNICIAN
EMPLOYEE
```

Asset Status

```text
AVAILABLE
ASSIGNED
MAINTENANCE
LOST
DISPOSED
```

Condition

```text
NEW
GOOD
FAIR
DAMAGED
SCRAP
```

Priority

```text
LOW
MEDIUM
HIGH
CRITICAL
```

Maintenance Status

```text
OPEN
IN_PROGRESS
COMPLETED
CANCELLED
```

---

7. Indexing Strategy

Indexes improve search performance.

Create indexes on

email

asset_code

serial_number

department_id

vendor_id

status

purchase_date

warranty_expiry

Example

```sql
CREATE INDEX idx_assets_status
ON assets(status);
```

---

8. Constraints

Unique

email

asset_code

serial_number

invoice_number

Foreign Keys

department_id

vendor_id

asset_id

user_id

Check Constraints

purchase_price > 0

repair_cost >= 0

expected_return >= assigned_date

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9. Audit Fields

Every important table contains

created_at

updated_at

deleted_at

This enables

- Tracking
- Recovery
- Compliance

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10. Soft Delete

Instead of deleting records

Set

deleted_at = CURRENT_TIMESTAMP

Advantages

- Recover data
- Preserve history
- Better audit support

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11. Sample Asset Record

```json
{
  "asset_code":"AST-000124",
  "asset_name":"Dell Latitude 7440",
  "category":"Laptop",
  "serial_number":"DL7440X992",
  "manufacturer":"Dell",
  "purchase_price":85000,
  "status":"ASSIGNED",
  "condition":"GOOD",
  "department_id":"uuid",
  "vendor_id":"uuid"
}
```

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12. Suggested Prisma Model (Example)

```prisma
model Asset {
  id            String   @id @default(uuid())
  assetCode     String   @unique
  assetName     String
  category      String
  serialNumber  String   @unique
```

```
  manufacturer    String
  purchasePrice   Decimal
  purchaseDate    DateTime
  warrantyExpiry  DateTime
  status          AssetStatus
  condition       AssetCondition
  qrCode          String?
  imageUrl        String?

  departmentId    String
  vendorId        String

  department Department @relation(fields: [departmentId], references: [id])
  vendor Vendor @relation(fields: [vendorId], references: [id])

  createdAt DateTime @default(now())
  updatedAt DateTime @updatedAt
  deletedAt DateTime?
}
```

---

13. Future Expansion

Additional tables planned

- asset_categories
- purchase_orders
- depreciation
- insurance

- warranties
- RFID_logs
- GPS_tracking
- IoT_sensor_data
- AI_predictions
- chatbot_history
- approval_workflows

Database Design Principles

✓ Fully Normalized (3NF)

✓ UUID Primary Keys

✓ Foreign Key Constraints

✓ Indexed Search Fields

✓ Soft Deletes

✓ Audit Logging

✓ AI-Compatible Structure

✓ Scalable for 100,000+ Assets

✓ Production Ready

✓ Prisma ORM Friendly